PRODUCT LINE PROFITABILITY & MARGIN PERFORMANCE ANALYSIS FOR NASSAU CANDY DISTRIBUTOR

Research Paper - Final Version

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ABSTRACT

This paper presents a comprehensive product-level profitability and margin performance analysis for Nassau Candy Distributor. Using transactional sales data, this study evaluates revenue, cost, and profit relationships across multiple product lines and divisions. The analysis focuses on identifying high-profit products, detecting margin inefficiencies, and understanding division-level financial performance.

Key findings reveal that a small percentage of products contribute to the majority of total profit, validating the Pareto principle. Certain high-sales products exhibit low profit margins due to elevated production costs. Division-wise analysis highlights variations in efficiency.

An interactive dashboard was developed using Streamlit to visualize these insights, enabling real-time filtering and decision-making.

Keywords: Profitability Analysis, Gross Margin, Data Analytics, Business Intelligence, Pareto Analysis, Cost Optimization

I. INTRODUCTION

In modern distribution businesses, revenue alone is not a sufficient indicator of success. Companies often experience situations where high sales volumes do not translate into proportional profitability due to rising costs and inefficiencies.

For organizations like Nassau Candy Distributor, understanding product-level profitability is essential for sustainable growth. Without proper analysis, decision-making related to pricing, sourcing, and product strategy remains reactive and intuition-driven.

This study aims to transform raw transactional data into actionable insights by evaluating product performance, division efficiency, and cost structures through data analytics techniques.

The application of Pareto analysis (80/20 rule) is widely used in profitability studies, where a small percentage of products contribute to the majority of profits. This concept is highly relevant in distribution businesses such as Nassau Candy.

II DATASET DESCRIPTION

The dataset used in this study consists of transactional sales records including product, division, and financial attributes.

Key Attributes:

- **Product Name**
- **Division (Chocolate, Sugar, Other)**
- **Sales**
- **Cost**
- **Gross Profit**
- **Units Sold**
- **Order Date**

The dataset enables detailed profitability analysis at both product and division levels.

Data quality checks confirmed:

- **No invalid sales or unit values**

- Proper date formatting
- Consistent financial calculations

TABLEL DATASET STRUCTURE SUMMARY

	Order Date	Division	Product Name	Sales	Cost	Gross Profit	Gross Margin (%)
0	2024-01-03 00:00:00	Chocolate	Wonka Bar - Milk Chocolate	6.5	2.28	4.22	64.9231
1	2024-01-04 00:00:00	Chocolate	Wonka Bar - Triple Dazzle Caramel	7.5	2.6	4.9	65.3333
2	2024-01-04 00:00:00	Chocolate	Wonka Bar - Nutty Crunch Surprise	10.47	3	7.47	71.3467
3	2024-01-04 00:00:00	Chocolate	Wonka Bar -Scrumdiddlyumptious	10.8	3.3	7.5	69.4444
4	2024-01-05 00:00:00	Chocolate	Wonka Bar - Triple Dazzle Caramel	11.25	3.9	7.35	65.3333
5	2024-01-06 00:00:00	Chocolate	Wonka Bar -Scrumdiddlyumptious	32.4	9.9	22.5	69.4444
6	2024-01-06 00:00:00	Chocolate	Wonka Bar - Triple Dazzle Caramel	7.5	2.6	4.9	65.3333
7	2024-01-06 00:00:00	Chocolate	Wonka Bar - Milk Chocolate	9.75	3.42	6.33	64.9231
8	2024-01-06 00:00:00	Chocolate	Wonka Bar - Nutty Crunch Surprise	6.98	2	4.98	71.3467
9	2024-01-06 00:00:00	Chocolate	Wonka Bar - Milk Chocolate	13	4.56	8.44	64.9231

III. METHODOLOGY

A. Data Cleaning

The dataset was preprocessed to ensure accuracy and reliability:

- Removed records with zero or negative sales
- Removed invalid unit entries
- Handled missing values using median imputation
- Converted date fields into standard format

B. KPI Calculation

The following key performance indicators were computed:

- Gross Margin (%) = (Gross Profit / Sales) × 100
- Profit per Unit = Gross Profit / Units
- Revenue Contribution
- Profit Contribution

Categories Identified:

- High Profit + High Margin (Top Performers)
- High Sales + Low Margin (Risk Products)
- Low Sales + Low Profit (Removal Candidates)

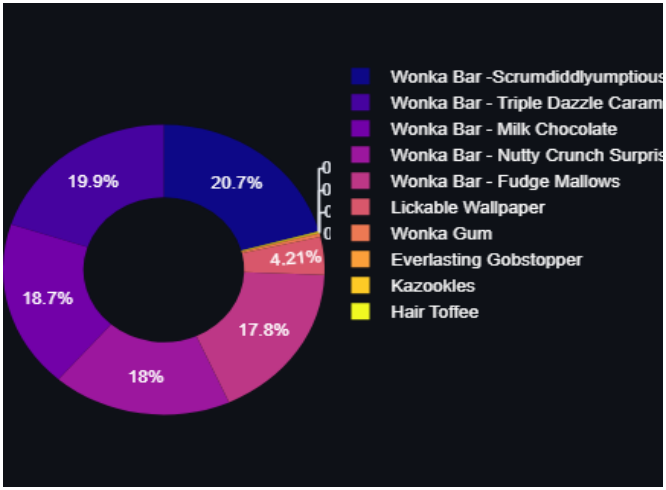

Fig. 1: *Product Profitability Distribution (Donut)*

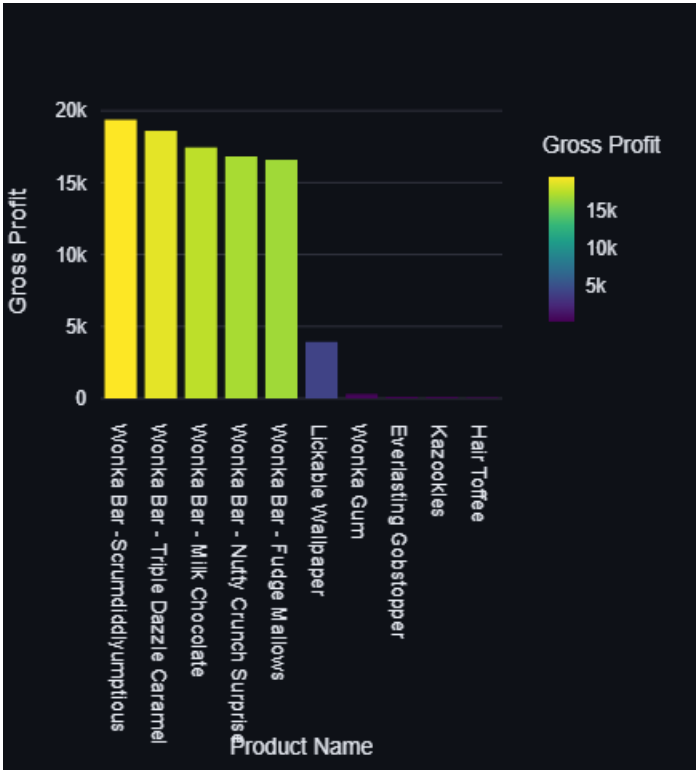

Fig. 2: *Product Profitability Distribution (Bar Chart)*

D. Division-Level Analysis

Performance comparison across divisions was conducted:

- Revenue vs Profit comparison
- Margin differences across divisions

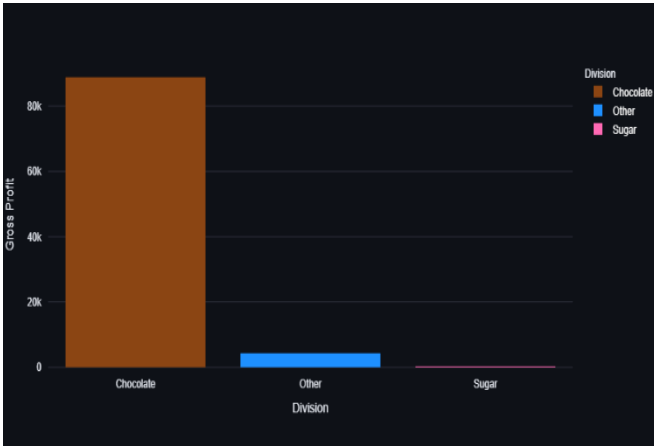

Fig. 3: Division-wise Profit Comparison (Bar Chart)

E. Cost vs Sales Diagnostics

A scatter analysis was performed to understand cost efficiency:

- High cost + low profit products identified
- Low cost + high profit products highlighted

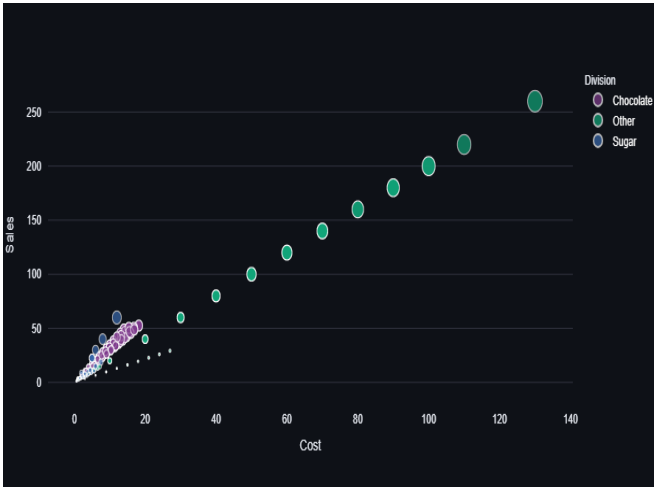

Fig. 4: Cost vs Sales Scatter Plot

F. Pareto Analysis (80/20 Rule)

The Pareto principle was applied to identify key profit-driving products:

- Top 20% products contribute to majority of profit
- Dependency risk identified

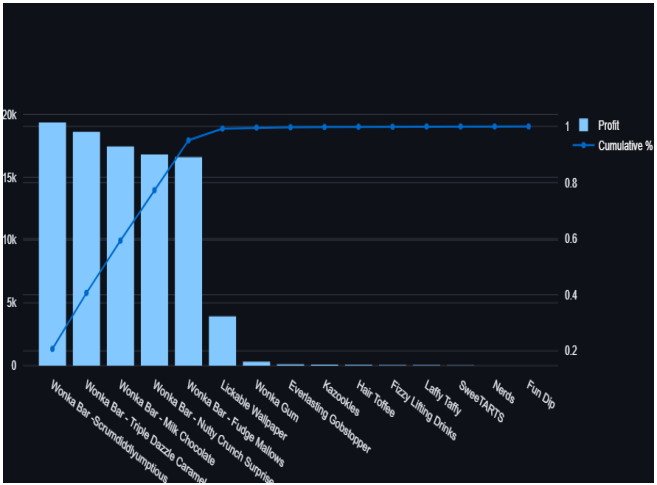

Fig. 5: Pareto Chart (Cumulative Profit Analysis)

G. Data Visualization Techniques

Various data visualization techniques were used to represent insights effectively:

- Donut Charts for product profit distribution
- Bar Charts for division-wise comparison
- Scatter Plots for cost vs sales analysis
- Pareto Charts for cumulative profit contribution

These visualizations help in identifying patterns, trends, and anomalies in the dataset.

H. Tools and Technologies Used

The following tools were used in this project:

- Python (Data Processing)
- Pandas (Data Cleaning and Analysis)
- Streamlit (Dashboard Development)
- Plotly (Interactive Visualization)

These tools enabled efficient data handling and interactive presentation of insights.

IV. KEY FINDINGS

- A small percentage of products generate the majority of profit
- Several high-revenue products have low profitability
- Division performance varies significantly across product categories
- Cost inefficiencies are present in certain product lines
- Profit concentration indicates dependency risk on limited products
- Some products with stable sales showed fluctuating margins over time
- Margin volatility indicates pricing or cost inconsistency
- Seasonal trends may impact sales and profit patterns.

V. STRATEGIC RECOMMENDATION

A. Optimize High-Cost Products

Products with high cost and low margin should be reviewed for cost reduction or repricing strategies.

B. **Focus on High-Profit Products**

Top-performing products should be prioritized for marketing and expansion.

C. Improve Division Efficiency

Divisions with low margins should optimize operations and sourcing strategies.

D. Reduce Dependency Risk

Diversify product portfolio to reduce reliance on limited high-profit items.

E. Pricing Strategy Enhancement

Adopt dynamic pricing models based on demand and cost fluctuations.

VI. KEY PERFORMANCE INDICATORS

KPI	Description
Gross Margin (%)	Profitability ratio
Profit per Unit	Efficiency per unit
Revenue Contribution	Share of total sales
Profit Contribution	Share of total profit

VII. LIMITATIONS OF THE STUDYS

This study has certain limitations:

- The dataset is limited to historical records and does not include real-time data
- External factors such as market trends and competition are not considered
- Cost breakdown is not detailed (fixed vs variable cost)
- The analysis is descriptive and does not include predictive modelling

Future studies can address these limitations by integrating advanced analytics and real-time data sources.

VII. CONCLUSION

This research demonstrates that high sales do not always equate to high profitability. By analyzing product-level margins and cost structures, businesses can gain deeper insights into financial performance.

The developed dashboard provides an interactive and scalable solution for monitoring profitability, enabling data-driven decision-making for pricing, cost optimization, and product strategy.

VIII. REFERENCES

[1] Unified Mentors Project Guidelines (2026)
[2] Python Documentation
[3] Pandas Library Documentation
[4] Streamlit Documentation
[5] Plotly Visualization Library